A counterexample to Erdős Problem 1075

Qiyuan Gu
University of Chicago
phoenix1203@uchicago.edu

Abstract

Erdős asked whether, for every fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$, every sufficiently large r-uniform hypergraph on N vertices with at least $(1+\varepsilon)(N/r)^r$ edges contains a subgraph on $m \to \infty$ vertices with at least $c_r m^r$ edges. We give counterexamples for every $r \geq 16$. The base construction is a sequence of explicit 16-uniform hypergraphs G_n whose unnormalized Lagrangians satisfy

$$16^{-16}\left(1+\frac{1}{4(14n)^{14}}\right) \leq \lambda(G_n) \leq 16^{-16} + \frac{1}{15!n}.$$

Blow-ups give the counterexample for $r = 16$, and adjoining common vertices to every edge lifts it to all larger uniformities. Thus Problem 1075, as a statement for all $r \geq 3$, has a negative answer. The endpoint question remains open for $3 \leq r \leq 15$, including the case 2/9 for 3-graphs.

1 Introduction

All hypergraphs in this paper are finite and simple. For a hypergraph H, write $v(H) = |V(H)|$, $e(H) = |E(H)|$, and let $H[S]$ denote the subgraph induced by $S \subseteq V(H)$.

Erdős proved that, for every fixed $r \geq 2$ and $\varepsilon > 0$, an r-uniform hypergraph on N vertices with at least εN^r edges contains a complete r-partite r-uniform hypergraph with equal part sizes tending to infinity with N [1]. In particular, it contains a subgraph on $m \to \infty$ vertices with $(m/r)^r$ edges. Erdős subsequently conjectured that, for each fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$ and all sufficiently large N, the assumption

$$e(H) \geq (1+\varepsilon)(N/r)^r$$

forces a subgraph on $m = m(N) \to \infty$ vertices with at least $c_r m^r$ edges [3, p. 80]. This is Problem 1075 in Bloom's collection [13].

With the usual density $d(H) = e(H)/\binom{v(H)}{r}$, the threshold $(N/r)^r$ corresponds asymptotically to $\alpha_r = r!/r^r$. Erdős proved that every number in $[0, \alpha_r)$ is a jump for r-graphs [2]. Thus Problem 1075 is the endpoint question at α_r. Frankl and Rödl disproved the earlier jumping constant conjecture by constructing non-jumps for every $r \geq 3$ [4]. Frankl, Peng, Rödl and Talbot later showed that $(5/2)r!/r^r$ is a non-jump for every $r \geq 3$ [5], and Peng proved the lifting theorem that carries a non-jump of the form $c\,r!/r^r$ to all larger uniformities [6]. Yan and Peng proved that 12/25 is a non-jump for 3-graphs, yielding $(54/25)r!/r^r$ for every $r \geq 3$ by lifting [7].

Shaw recently proved that $2r!/r^r$ is a non-jump for every $r \geq 4$, and that for $r \geq 4$ no smaller non-jump can be obtained from his finite-pattern formulation of the Frankl–Rödl method [8, Theorems 4.2 and 4.3]. Liu and Mubayi subsequently proved that 4/9 is a non-jump for 3-graphs by a construction outside that finite-pattern framework [9]. The endpoint 2/9 for 3-graphs remains open; Baber and Talbot proved, using flag algebras, the first jump intervals immediately above it [10]. The result below shows that the endpoint is a non-jump for every $r \geq 16$; the cases $3 \leq r \leq 15$ remain open.

The construction below is not an instance of Shaw's finite-pattern criterion. Its distinguished links between the two vertex families follow the matching A_iB_i and the shifted matching B_iA_{i+1} around a cycle, and the proof controls the full Lagrangian of a growing sequence G_n. Hence Shaw's Theorem 4.3 does not apply.

Theorem 1. *For every $\gamma > 16^{-16}$, there is an $\varepsilon > 0$ and there are arbitrarily large 16-uniform hypergraphs H such that*

$$e(H) \geq (1+\varepsilon)(v(H)/16)^{16}, \qquad e(H[S]) < \gamma|S|^{16} \quad \textit{for every nonempty } S \subseteq V(H).$$

The conclusion is stronger than the formulation of Problem 1075, since the bound holds for every nonempty vertex subset. Taking $16^{-16} < \gamma < c_{16}$ in Theorem 1 rules out any $c_{16} > 16^{-16}$. In the usual density normalization, it also implies that $16!/16^{16}$ is a non-jump. The proof uses explicit finite hypergraphs whose Lagrangians approach 16^{-16} from above. We use the standard unnormalized Lagrangian [11, 12],

$$P_H(x) = \sum_{e \in E(H)} \prod_{v \in e} x_v, \qquad \lambda(H) = \max_{\substack{x_v \geq 0 \\ \sum_v x_v = 1}} P_H(x), \tag{1.1}$$

without the factor $r!$.

Proposition 2. *For every integer $n \geq 2$, the 16-uniform hypergraph G_n defined in Section 2 has $2n+29$ vertices and satisfies*

$$16^{-16}\left(1 + \frac{1}{4(14n)^{14}}\right) \leq \lambda(G_n) \leq 16^{-16} + \frac{1}{15!n}. \tag{1.2}$$

The mechanism is visible at the level of the cubic forms. After normalizing the bottom mass, the balanced weighting on the closed cycle has

$$b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{3},$$

so both cubic forms attain $1/27$ even with $D > 0$, while the additional edge $\{U, V\} \cup \mathcal{D}$ contributes a gain of order D^{14}. Deleting one pair A_k, B_k opens the cycle; the terminal mismatch forces a loss of order D^3, which dominates that gain. Consequently every open path T_L has Lagrangian exactly 16^{-16}, whereas the closed cycle has Lagrangian slightly above 16^{-16}.

2 The cyclic construction

Fix $n \geq 2$. Take pairwise distinct vertices

$$A_i, B_i \quad (i \in \mathbb{Z}/n\mathbb{Z}), \qquad C, U, V, \qquad W_1, \dots, W_{12}, \qquad D_1, \dots, D_{14}.$$

Put $\mathcal{W} = \{W_1, \dots, W_{12}\}$ and $\mathcal{D} = \{D_1, \dots, D_{14}\}$. On the vertex set $\{A_i, B_i : i \in \mathbb{Z}/n\mathbb{Z}\} \cup \{C\} \cup \mathcal{D}$, define two three-uniform hypergraphs by

$$E(H_1) = \bigcup_i \{\{A_i, X, Y\} : X \in \{B_i, C\},\ Y \in \{B_j : j \neq i\} \cup \mathcal{D}\},$$
$$E(H_2) = \bigcup_i \{\{B_i, X, Y\} : X \in \{A_{i+1}, C\},\ Y \in \{A_j : j \neq i+1\} \cup \mathcal{D}\}.$$

All subscripts are read modulo n. Define

$$\begin{aligned} E(G_n) = \{&\mathcal{W} \cup \{U\} \cup e : e \in E(H_1)\} \\ &\cup \{\mathcal{W} \cup \{V\} \cup e : e \in E(H_2)\} \cup \{\{U, V\} \cup \mathcal{D}\}. \end{aligned} \tag{2.1}$$

Thus G_n is a simple 16-uniform hypergraph.

Use lower-case letters for the weights of the corresponding vertices, and write

$$A = \sum_i a_i, \qquad B = \sum_i b_i, \qquad D = \sum_{j=1}^{14} d_j, \qquad \omega = \prod_{j=1}^{12} w_j.$$

The two cubic forms and the polynomial of G_n are

$$P_1 = \sum_i a_i(b_i + c)(B - b_i + D), \tag{2.2}$$

$$P_2 = \sum_i b_i(a_{i+1} + c)(A - a_{i+1} + D), \tag{2.3}$$

$$P_{G_n} = \omega(uP_1 + vP_2) + uv \prod_{j=1}^{14} d_j. \tag{2.4}$$

Lemma 3. *The lower bound in* (1.2) *holds.*

Proof. Assign the weights

$$\begin{aligned} w_j = \frac{1}{16}, \qquad u = v = \frac{1}{32}, \qquad a_i = b_i = \frac{1}{16n}, \\ d_j = \frac{1}{224n}, \qquad c = \frac{n-1}{16n}. \end{aligned} \tag{2.5}$$

Their sum is one. Moreover,

$$A = B = c + D = \frac{1}{16}, \qquad b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{16}.$$

Thus $P_1 = P_2 = 16^{-3}$. The first term in (2.4) is 16^{-16}, and the second is

$$\frac{1}{1024(224n)^{14}} = \frac{16^{-16}}{4(14n)^{14}}.$$

□

3 A bound for open paths

We first prove an inequality for the cubic forms obtained by opening the cycle.

Lemma 4. *Let $L \geq 1$, and let $a_1, \ldots, a_L, b_1, \ldots, b_L, c, D$ be nonnegative numbers. Put*

$$A = \sum_{i=1}^{L} a_i, \qquad B = \sum_{i=1}^{L} b_i, \qquad a_{L+1} = 0,$$

and suppose that $A + B + c + D = 1$. Define P_1, P_2 by (2.2) *and* (2.3)*, with sums over $1 \leq i \leq L$. Then, for every $s \in [0, 1]$,*

$$sP_1 + (1 - s)P_2 \leq \frac{1}{27} - \frac{s(1-s)D^3}{1200}. \tag{3.1}$$

Proof. Set

$$\delta_A = A - \frac{1}{3}, \qquad \delta_B = B - \frac{1}{3},$$

and

$$\begin{aligned}
\beta &= \frac{B + D - c}{2} = D + \frac{\delta_A}{2} + \delta_B, \\
\alpha &= \frac{A + D - c}{2} = D + \delta_A + \frac{\delta_B}{2}.
\end{aligned}$$

Completing the square gives

$$P_1 = \frac{A(1-A)^2}{4} - \sum_i a_i (b_i - \beta)^2, \tag{3.2}$$

$$P_2 = \frac{B(1-B)^2}{4} - \sum_i b_i (a_{i+1} - \alpha)^2. \tag{3.3}$$

If

$$E = s \sum_i a_i (b_i - \beta)^2 + (1-s) \sum_i b_i (a_{i+1} - \alpha)^2,$$

then (3.2)–(3.3) give the exact identity

$$\begin{aligned}
\frac{1}{27} - sP_1 - (1-s)P_2 &= \frac{s\delta_A^2(4/3 - A)}{4} + \frac{(1-s)\delta_B^2(4/3 - B)}{4} + E \\
&\geq \frac{s\delta_A^2 + (1-s)\delta_B^2}{12} + E,
\end{aligned} \tag{3.4}$$

since $0 \leq A, B \leq 1$.

Case 1: $D = 0$ or $s \in \{0, 1\}$. The right side of (3.1) is then $1/27$, so the claim follows immediately from (3.4).

Case 2: one of $|\delta_A|, |\delta_B|$ is at least $D/10$. Assume first that $|\delta_A| \geq D/10$. Since $D \leq 1$,

$$\frac{s\delta_A^2}{12} \geq \frac{sD^2}{1200} \geq \frac{s(1-s)D^3}{1200}.$$

The case $|\delta_B| \geq D/10$ is the same.

Case 3: $|\delta_A|, |\delta_B| < D/10$. Here

$$D \leq c + D = \frac{1}{3} - \delta_A - \delta_B \leq \frac{1}{3} + \frac{D}{5},$$

so $D \leq 5/12$. It follows that

$$\frac{17D}{20} \leq \alpha, \beta \leq \frac{23D}{20} \leq \frac{23}{48}, \qquad A, B \geq \frac{1}{3} - \frac{D}{10} \geq \frac{7}{24}. \tag{3.5}$$

In particular, $\alpha, \beta > 0$, $A \geq \alpha/3$, and $B \geq \beta/3$.

Consider the finite sequence

$$\frac{a_1}{\alpha}, \frac{b_1}{\beta}, \ldots, \frac{a_L}{\alpha}, \frac{b_L}{\beta}, 0. \tag{3.6}$$

Each successive pair z, z' contributes to E a term

$$s\alpha\beta^2 z(z'-1)^2 \quad \text{or} \quad (1-s)\beta\alpha^2 z(z'-1)^2,$$

according as the first entry comes from an a_i or a b_i. *Case 3a: some entry of* (3.6) *is at least* $1/3$. Because the last entry is zero, some successive pair satisfies $z \geq 1/3$ and $z' < 1/3$. Hence

$$E \geq \frac{4}{27} \min\{s\alpha\beta^2, (1-s)\beta\alpha^2\}. \tag{3.7}$$

Case 3b: all entries of (3.6) *are less than* $1/3$. Then

$$\begin{aligned} E &\geq \frac{4}{9}(s\beta^2 A + (1-s)\alpha^2 B) \\ &\geq \frac{4}{27}(s\alpha\beta^2 + (1-s)\beta\alpha^2), \end{aligned}$$

which also implies (3.7). Using (3.5), we obtain in either case

$$E \geq \frac{4}{27}\left(\frac{17}{20}\right)^3 \min\{s, 1-s\} D^3 \geq \frac{s(1-s)D^3}{1200}.$$

The last inequality follows from $\frac{4}{27}(\frac{17}{20})^3 = 4913/54000 > 1/1200$. Together with (3.4), this proves the lemma. □

For $L \geq 1$, define the 16-uniform hypergraph T_L as in (2.1), with each index ranging over $1, \ldots, L$ and with the last link in H_2 replaced by the triples

$$\{B_L, C, Y\}, \qquad Y \in \{A_1, \ldots, A_L\} \cup \mathcal{D}.$$

The edge $\{U, V\} \cup \mathcal{D}$ is retained. Its polynomial is (2.4) with $a_{L+1} = 0$.

Lemma 5. *For every $L \geq 1$, we have $\lambda(T_L) = 16^{-16}$.*

Proof. The lower bound follows by assigning weight 1/16 to the vertices of any one edge and zero elsewhere. For the upper bound, take nonnegative vertex weights of total mass one, and put

$$h = \sum_{j=1}^{12} w_j, \qquad q = u + v, \qquad t = A + B + c + D.$$

Thus $h + q + t = 1$. If $q = 0$ or $t = 0$, then $P_{T_L} = 0$. Otherwise put

$$s = u/q, \qquad d = D/t, \qquad z = s(1-s)d^3 \in [0, 1/4].$$

Applying Lemma 4 to the bottom weights divided by t, and then using homogeneity, gives

$$uP_1 + vP_2 \leq qt^3\left(\frac{1}{27} - \frac{z}{1200}\right).$$

The arithmetic–geometric mean inequality now yields

$$P_{T_L} \leq \left(\frac{h}{12}\right)^{12} qt^3\left(\frac{1}{27} - \frac{z}{1200}\right) + q^2 s(1-s)\left(\frac{td}{14}\right)^{14}. \tag{3.8}$$

Also,

$$\frac{qt^3}{27} \leq \left(\frac{q+t}{4}\right)^4, \qquad \frac{q^2 t^{14}}{14^{14}} \leq 4\left(\frac{q+t}{16}\right)^{16}. \tag{3.9}$$

Define

$$F(h) = \left(\frac{4h}{3}\right)^{12} [4(1-h)]^4.$$

We have $F(h) \leq 1$, with equality at $h = 3/4$. Since $1 - 27z/1200 > 0$, we may substitute (3.9) into (3.8). Using $d^{14} \leq d^3$, we get

$$\begin{aligned} \frac{P_{T_L}}{16^{-16}} &\leq F(h)\left(1 - \frac{9}{400}z\right) + 4s(1-s)d^{14}(1-h)^{16} \\ &\leq F(h) + z\left[4(1-h)^{16} - \frac{9}{400}F(h)\right]. \end{aligned} \tag{3.10}$$

Suppose first that $1/2 \leq h < 1$. Then

$$\frac{(1-h)^{16}}{F(h)} = \frac{(3/4)^{12}}{256}\left(\frac{1-h}{h}\right)^{12} \leq \frac{1}{256}.$$

Since $1/64 < 9/400$, the expression in brackets in (3.10) is nonpositive. Thus the right side is at most $F(h) \leq 1$. If $h = 1$, the polynomial is zero.

Suppose next that $0 \leq h \leq 1/2$. Discarding the negative term in (3.10) and using $z \leq 1/4$, we obtain

$$\frac{P_{T_L}}{16^{-16}} \leq F(h) + (1-h)^{16}. \tag{3.11}$$

For $0 < h \leq 1/2$, the function $F(h)/h$ is increasing, since its logarithmic derivative is

$$\frac{11}{h} - \frac{4}{1-h} = \frac{11 - 15h}{h(1-h)} > 0.$$

Moreover,

$$\frac{F(1/2)}{1/2} = 32\left(\frac{2}{3}\right)^{12} = \frac{131072}{531441} < 1.$$

It follows that $F(h) \leq h$ throughout $[0, 1/2]$. The right side of (3.11) is therefore at most $h + (1-h)^{16} \leq 1$, as required. $\square$

4 The Lagrangian bound and the blow-ups

Proof of Proposition 2. The lower bound was proved in Lemma 3. For the upper bound, take any nonnegative weights on G_n of total mass one. Choose an index k with

$$a_k + b_k \leq \frac{A+B}{n} \leq \frac{1}{n}.$$

Put $a'_j = a_{k+j}$, $b'_j = b_{k+j}$ for $1 \leq j \leq n-1$, with indices modulo n, and set $A' = \sum_j a'_j$, $B' = \sum_j b'_j$, $a'_n = 0$. After deleting A_k, B_k, the remaining polynomial is

$$\begin{aligned} P_{G_n - \{A_k, B_k\}} &= \omega u \sum_{j=1}^{n-1} a'_j(b'_j + c)(B' - b'_j + D) \\ &\quad + \omega v \sum_{j=1}^{n-1} b'_j(a'_{j+1} + c)(A' - a'_{j+1} + D) + uv\prod_{\ell=1}^{14} d_\ell \\ &= P_{T_{n-1}}. \end{aligned} \tag{4.1}$$

Thus Lemma 5 and homogeneity bound the total weight of the edges that remain by

$$16^{-16}(1-a_k-b_k)^{16} \le 16^{-16}.$$

For any simple 16-uniform hypergraph and weights of total mass one,

$$\frac{\partial P_H}{\partial x_v} \le \sum_{\substack{S\subseteq V(H)\setminus\{v\}\\ |S|=15}} \prod_{w\in S} x_w \le \frac{\left(\sum_{w\ne v} x_w\right)^{15}}{15!} \le \frac{1}{15!}. \tag{4.2}$$

Indeed, each product of fifteen distinct weights occurs 15! times in the power expansion, and all other terms are nonnegative. The deleted edges consequently have total weight at most

$$a_k\frac{\partial P_{G_n}}{\partial a_k} + b_k\frac{\partial P_{G_n}}{\partial b_k} \le \frac{a_k+b_k}{15!} \le \frac{1}{15!n}.$$

This proves (1.2). □

Proof of Theorem 1. Fix $\gamma > 16^{-16}$, and choose an integer $n \ge 2$ such that

$$16^{-16} + \frac{1}{15!n} < \gamma.$$

Keep n fixed, and set

$$\varepsilon = \frac{1}{4(14n)^{14}} > 0.$$

For each positive integer M, form a blow-up $B_{n,M}$ of G_n: replace each vertex by an independent class, and each edge by all transversal edges across its sixteen classes. Use the following class sizes, with one class for each listed vertex:

vertex	W_j	U, V	A_i, B_i	C	D_j
class size	$14nM$	$7nM$	$14M$	$14(n-1)M$	M

There are $N = 224nM$ vertices. These class sizes are proportional to the weights in (2.5), and the computation in Lemma 3 gives the exact edge count

$$e(B_{n,M}) = \left[(14n)^{16} + 49n^2\right]M^{16} = (1+\varepsilon)(N/16)^{16}. \tag{4.3}$$

Let S be any nonempty set of m vertices of $B_{n,M}$, and let k_v be the number of its vertices in the class corresponding to $v \in V(G_n)$. Then

$$e(B_{n,M}[S]) = \sum_{e\in E(G_n)} \prod_{v\in e} k_v = m^{16} P_{G_n}((k_v/m)_v) \le m^{16}\lambda(G_n) < \gamma m^{16}.$$

Together with (4.3) and $M \to \infty$, this proves the theorem. A non-induced subgraph has no more edges than the induced subgraph on the same vertex set. □

Corollary 6. *For every integer $r \ge 16$ and every $\gamma > r^{-r}$, there is an $\varepsilon > 0$ and there are arbitrarily large r-uniform hypergraphs H such that*

$$e(H) \ge (1+\varepsilon)(v(H)/r)^r, \qquad e(H[S]) < \gamma|S|^r \quad \textit{for every nonempty } S \subseteq V(H).$$

In particular, Problem 1075 fails for every $r \ge 16$.

Proof. This is the lifting step of Peng [6]; in the present normalization it has the following direct form. Let $q = r - 16$. For a 16-uniform hypergraph G, form an r-uniform hypergraph G^{+q} by adjoining the same set of q new vertices to every edge. If the total weight on $V(G)$ is t, then the arithmetic–geometric mean inequality gives

$$P_{G^{+q}} \le \lambda(G) t^{16} \left(\frac{1-t}{q}\right)^q,$$

with the usual interpretation when $q = 0$. The right side is maximized at $t = 16/r$, and equality is attained by taking an optimal weighting of G, scaled by $16/r$, and weight $1/r$ on each new vertex. Hence

$$\lambda(G^{+q}) = \frac{16^{16}}{r^r} \lambda(G). \tag{4.4}$$

Applying (4.4) to G_n and using Proposition 2 gives

$$r^{-r}\left(1 + \frac{1}{4(14n)^{14}}\right) \le \lambda(G_n^{+q}) \le r^{-r} + \frac{16^{16}}{r^r 15! n}.$$

Choose n so that the upper bound is less than γ. The blow-up argument in the proof of Theorem 1, using the rational weighting obtained by suspending (2.5), then gives the stated r-uniform hypergraphs with $\varepsilon = 1/[4(14n)^{14}]$. $\square$

Declaration of generative AI and AI-assisted technologies

GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition. GPT-6 Astra was run in a research environment containing earlier results produced by GPT-5.6 Sol and Claude Opus 5, but those earlier results did not contribute to the final mathematical arguments. The author reviewed the final manuscript and takes full responsibility for its content.

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